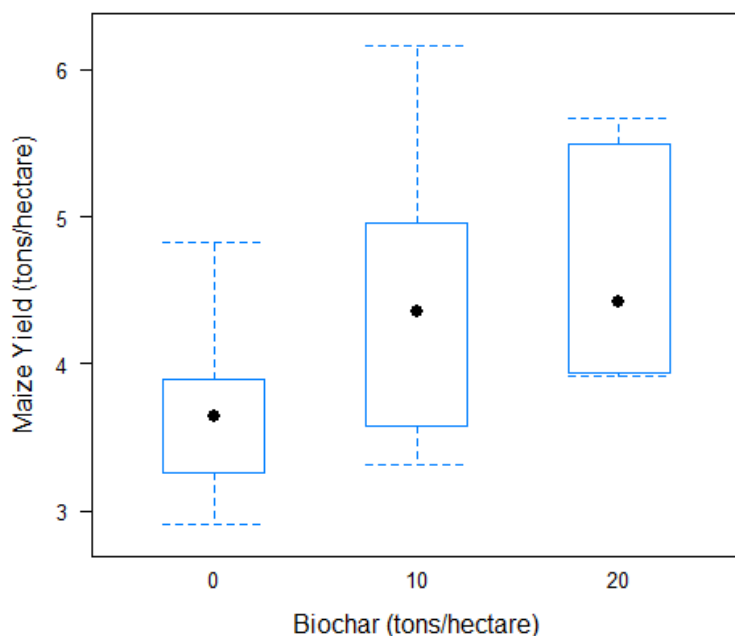


Question 1

Biochar is the carbonaceous product from the burning of biomass. When added to the soil, biochar has been shown to improve the soil's chemical properties. A study was conducted in Nepal to see if biochar could also be used to improve the grain yield from maize crops. An invasive forest shrub was used as feedstock to produce the biochar used in the study. The field trial was conducted on private rain-fed agricultural land which was divided into twenty-four plots of 10m² each. Treatments of 0, 10 and 20 tons/hectare of biochar with eight replications each were applied in a completely randomised design to the plots. In addition to the biochar treatment, equal amounts of mineral fertiliser and farmyard manure was applied to each plot according to standard farming practice. The following plot shows a summary of the results:



The following table shows the corresponding summary statistics:

Treatment	n	$\bar{x}$	s
0 t/ha (control)	8	3.67	0.59
10 t/ha	8	4.4	0.97
20 t/ha	8	4.66	0.77

- a) Briefly describe the distribution of the maize yield from plots treated with 10 tons/hectare of biochar.

The distribution seems slightly skewed to the right.

- b) Researchers were interested in testing if biochar affects the mean yield from maize crops. State the null hypothesis in words for this study.

H_0 : mean yield is the same for each biochar group

- c) An analysis of variance was conducted to compare the mean maize yield for the three treatment levels of biochar. Ignoring the treatment groups, the sample standard deviation for the yields was 0.87. Based on this and the values given in the previous table, complete the following ANOVA table.

Source	DF	SS	MS	F
Treatment Groups	2	4.24	2.12	3.38
Residuals	21	13.17	0.627	
Total	23	17.41		

Note $s = \sqrt{\frac{\sum (x_j - \bar{x})^2}{n-1}}$ so $\sum (x_j - \bar{x})^2 = (n-1)s^2$

Here Total SS = $(24-1) \times 0.87^2 = 17.41$

Similarly, Residual SS = $(8-1) \times 0.59^2 + (8-1) \times 0.97^2 + (8-1) \times 0.77^2$
 $= 13.17$

- d) Is there significant evidence of a difference in mean maize yield across the three treatment levels of biochar at the 10% level? Justify your conclusion.

P-value is $P(F_{2,21} > 3.38) = .053$, $\leftarrow 1 - pf(3.38, 2, 21)$
 so there is evidence of a difference in mean yield at the .1 level.

- e) Tukey's Honest Significant Differences were calculated in R. The output is given below. Based on this output, which pairwise comparisons, if any, are significantly different at the 10% level? Why do we use Tukey's Honest Significant Difference to do pairwise comparisons of treatments instead of multiple two sample t-tests at the 10% level?

```
Tukey multiple comparisons of means
90% family-wise confidence level

Fit: aov(formula = yield ~ treatment, data = farm)

$treatment
      diff      lwr      upr    p adj
10-0  0.7343754 -0.1279105  1.596661 0.1787670
20-0  0.9919266  0.1296407  1.854213 0.0525616
20-10 0.2575512 -0.6047348  1.119837 0.7954369
```

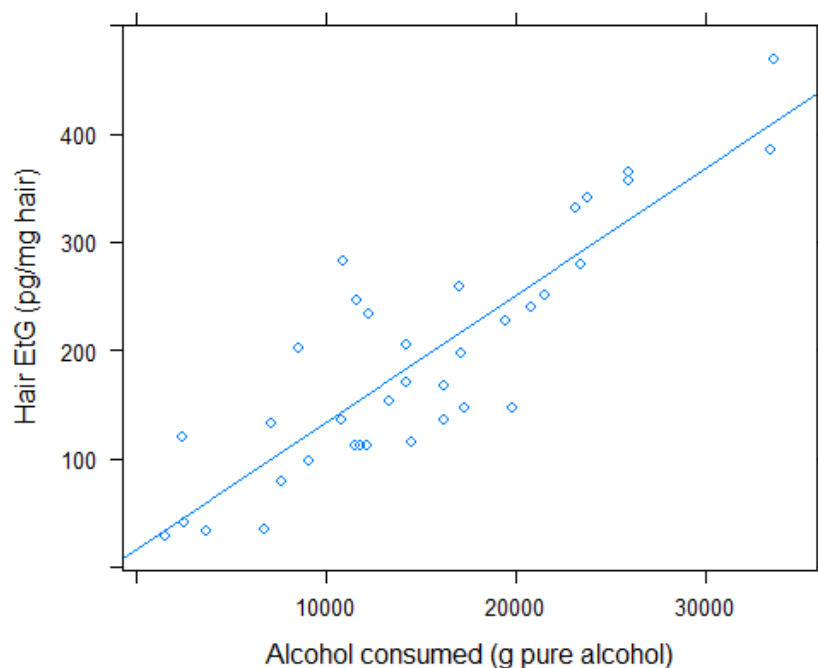
- There is a significant difference in mean yield between the 0 ton/hectare and 20 ton/hectare treatments.
- Doing multiple two-sample t-tests will inflate the Type I error rate.

- f) How might the assumptions underlying the one-way analysis of variance be violated in this study?

Assumptions of constant and Normal variability seem fine from the box plots.
 Could be issues with independence between neighbouring plots.

Question 2

Ethyl glucuronide (EtG), an alcohol metabolite, is known to accumulate in hair. A study was undertaken to determine its suitability as a marker of chronic and excessive alcohol consumption. The study involved 36 patients entering a detoxification program. Hair samples (first 3 cm segments) were taken from each patient and the EtG level measured (units pg/mg hair) by gas chromatography coupled to mass spectrometry. Interviews were conducted with patients to determine alcohol consumption (units: g pure alcohol) over the past three months. The data are displayed in the figure below together with the fitted least squares line.



The output on the next page show the results of a linear regression fit in R for the relationship between EtG levels in hair and past alcohol consumption:

```
lm(formula = hair ~ alcohol)

Residuals:
    Min       1Q   Median       3Q      Max
-101.97  -30.04  -15.74   43.16  138.28

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 15.872784   19.611596   0.809    0.424
alcohol      0.011776    0.001155  10.195 7.09e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 54.69 on 34 degrees of freedom
Multiple R-squared:  0.7535, Adjusted R-squared:  0.7463
F-statistic: 103.9 on 1 and 34 DF, p-value: 7.092e-12
```

- a) Briefly interpret the values 15.872784 and 0.011776 in the regression output.

A person with no alcohol consumption would have 15.87 pg/mg EtG in their hair on average.

A 1 g increase in alcohol consumption is associated with a mean increase of 0.0118 pg/mg EtG in hair.

- b) Give a 95% confidence interval for the underlying slope of the linear relationship between the level of EtG in hair and alcohol consumption.

95% CI is $0.011776 \pm 2.032 \times 0.001155$

$\downarrow$ $qt(.975, 34)$

$= 0.0118 \pm 0.0024$

- c) Does the regression analysis provide evidence that mean EtG concentration in hair will be different to zero when a patient's past alcohol consumption is zero? State the null and alternative hypotheses, and report the appropriate test statistic and P-value from the output. What do you conclude?

Let β_0 = mean EtG concentration in hair for a person with 0 alcohol consumption.

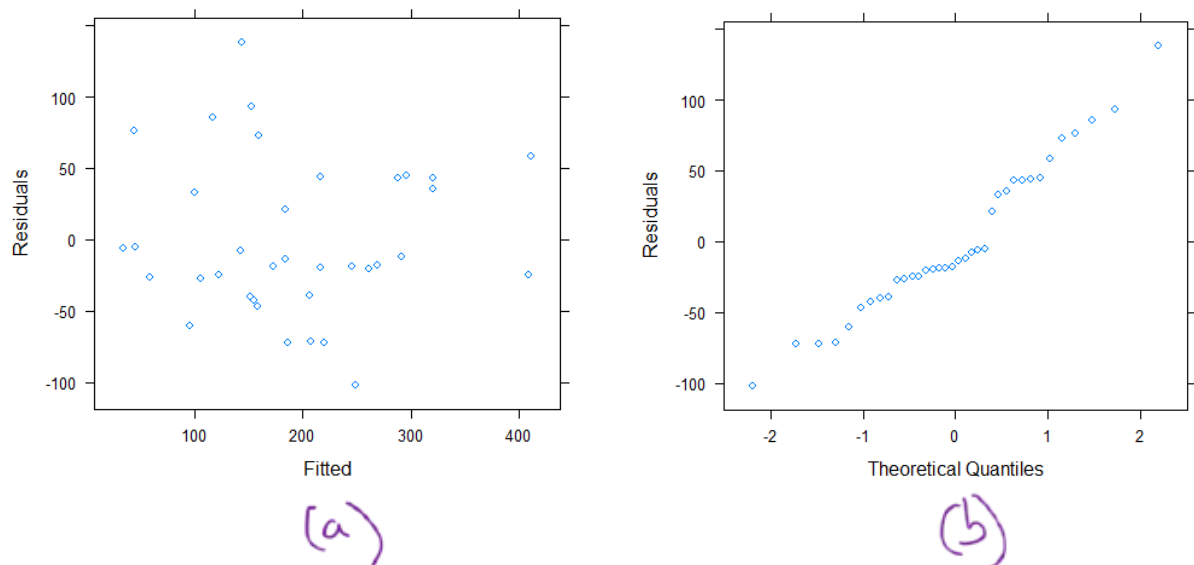
Test $H_0: \beta_0 = 0$ vs $H_1: \beta_0 \neq 0$,
using $t_{34} = 0.809$.

Reported p-value is 0.424, inconclusive evidence that the mean concentration differs from 0.

- d) A patient whose past alcohol consumption was 19800 g pure alcohol had EtG level of 145 pg/mg hair in their hair sample. What is the residual associated with this patient in the regression model.

$$\begin{aligned}\text{Residual} &= \text{observed} - \text{predicted} \\ &= 145 - (15.8727 + 0.011776 \times 19800) \\ &\approx 145 - 249 = -104 \text{ pg/mg}.\end{aligned}$$

- e) The following figures were generated by R to help check the assumptions underlying the linear regression. The theoretical quantiles on the horizontal axis of the plot on the left are those of the standard normal distribution.



Comment on the validity of the assumptions underlying linear regression for this data, with reference to these figures and the figure on the first page of this question.

Figure (a) shows generally constant variability of residuals and suggests linearity assumption is reasonable since there is no strong pattern remaining.

The points in (b) are close to a straight line, suggesting the Normality assumption is reasonable. (However, there are potentially two distributions, as shown)

Question 3

Resting heart rate (RHR) and maximal oxygen uptake (VO2Max) are commonly used indicators of cardiovascular fitness. Individuals with higher aerobic fitness generally have a greater VO2Max and may also have a lower risk of developing cardiovascular conditions such as hypertension.

A sports science research team conducted a study involving 320 adults. For each participant, the resting heart rate (RHR), maximal oxygen uptake (VO2Max), and whether the participant had hypertension were recorded.

- a) In this study, the Pearson correlation coefficient between VO2Max and RHR was found to be $r = -0.58$. Construct a 95% confidence interval for the population correlation coefficient ρ between VO2Max and RHR

The confidence interval from Z:

$$\begin{aligned} & \frac{1}{2} \ln \left(\frac{1+r}{1-r} \right) \pm z_{0.975}^* \times se(z) \\ &= \frac{1}{2} \ln \left(\frac{1+(-0.58)}{1-(-0.58)} \right) \pm 1.96 \times \frac{1}{\sqrt{317}} \\ &= -0.6625 \pm 0.1101 \\ &= (-0.7725, -0.5524) \end{aligned}$$

We then transform the bounds back to correlation values to obtain the CI for ρ :
 $(\tanh(-0.7725), \tanh(-0.5524)) = (-0.6484, -0.5023)$

- b) To investigate the relationship between aerobic fitness and hypertension, the following logistic regression model was fitted:

```
glm(Hypertension ~ VO2Max, data = fitness, family = "binomial")
```

The relevant R output is shown below.

Coefficients:		
	Estimate	Std. Error
(Intercept)	3.75440	0.68868
VO2Max	-0.08830	0.01677

Note that "Hypertension" = 1 if the participant has hypertension, and 0 otherwise.

Write down the fitted logistic regression model relating the probability of hypertension to VO2Max.

Let $p = P(\text{Hypertension})$.

The fitted model is

$$\ln\left(\frac{p}{1-p}\right) = 3.7544 - 0.0833 \text{ VO2Max}.$$

Equivalently, we can write

$$p = \frac{e^{3.7544 - 0.0833 \text{ VO2Max}}}{1 + e^{3.7544 - 0.0833 \text{ VO2Max}}}.$$

- c) Estimate the probability that a participant with VO2Max = 40 has hypertension.

$$\text{logit}(\hat{p}) = 3.7544 - 0.0833 \times 40 = 0.2224.$$

$$\hat{p} = \frac{e^{0.2224}}{1 + e^{0.2224}} \approx 0.555.$$

Therefore the estimated probability is about 55.5%.

- d) Construct a 95% confidence interval for the odds ratio associated with a 1-unit increase in VO2Max, and interpret the result.

95% CI for β_1 :

$$-0.0883 \pm 1.96 \times 0.01677 = (-0.1212, -0.0554)$$

Therefore the 95% CI for the odds ratio is

$$(e^{-0.1212}, e^{-0.0554}) = (0.886, 0.946).$$

As the interval lies entirely below 1, this suggests that higher VO2Max is associated with lower odds of hypertension.